Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Spring-2019

Program: B.Sc. in CSE (Batch: 8^{th}) 1^{st} Year 1^{st} Semester

Course Code: PHY 1101

Course Title: Physics – I

Time: 1 hour & 30 minutes | Total Marks: 30

Group-A

[Answer any 5 (Five) questions including No. 1.] **5 × 5 = 25**

1. Read the stem carefully and answer the questions that follows –

A capacitor consisting of two concentric spheres of radii 30 and 31 cm with 500V potential difference.

a) Define dielectric constant. 1

b) How much charge is stored in the capacitor? 3

c) What will be happened by using the dielectric material in the above capacitor? 1

2 a) Give an idea of Gaussian surface. 1

b) Find the net flux of a cylindrical Gaussian surface placed in an external electric field. 4

3. a) Write down the relation between current density & drift velocity. 1

b) A silver wire 1mm in diameter carries a charge of 90 coulomb in 1 hr. Silver contains 5.8×10^{28} free electrons/m^3. Calculate –

(i) the current in the wire 2

(ii) current density and drift velocity of electrons in the wire. 2

4. a) Draw the lines of force for the positive and negative charges. 2

b) Two charges $q_1 = 1.0 \times 10^{-6}$ coul and $q_2 = 2.0 \times 10^{-6}$ coul are kept 10 cm away from each other. At what point the line joining the two charges is the electric field strength zero? 3

5. a) Define electric field from potential gradient. 1

b) What is the potential at the surface of an aluminium nucleus? Given the radius of the nucleus, $r = 4.5 \times 10^{-13}$cm and its charge $q = 13e$. Where e is the charge of the electron. (Assume that, this charge is concentrated at the center of the aluminium nucleus) 4

6. a) Give some limitation of the Coulomb's law. 1

b) Write a short note on – : (Any two) 2+2=4

i) Quantization of electric charge

ii) Equipotential surface

iii) Specific permittivity

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Time: 10 min — **[Answer all questions]** — **0.5 × 10 = 5**

Student Name:	**ID No.:**

[Tick (√) mark the appropriate one]

1. 1 Farad is –
   - ☐ a) 1 Coulomb per 1 Newton
   - ☐ b) 1 Coulomb per 1 Joule
   - ☐ c) 1 Coulomb per 1 Volt
   - ☐ d) 1 Volt per 1 Coulomb
   - ☐ e) 1 Newton per 1 Coulomb

2. The reciprocal of resistivity is called the –
   - ☐ a) resistivity
   - ☐ b) specific resistance
   - ☐ c) conductance
   - ☐ d) conductivity
   - ☐ e) permittivity

3. With an equality or balance of charge, the object said to be-
   - ☐ a) maximum charged
   - ☐ b) negatively charged
   - ☐ c) positively charged
   - ☐ d) electrically neutral
   - ☐ e) mixed charged

4. If the surface dose not enclosed by the charge, then the flux $\oint_s \bar{E}.\overline{da} = ?$
   - ☐ a) q
   - ☐ b) $\frac{q}{\epsilon_0}$
   - ☐ c) $\epsilon_0 q$
   - ☐ d) $\frac{\epsilon_0}{q}$
   - ☐ e) 0

5. The potential differences between two points in an equipotential surface is -
   - ☐ a) 0
   - ☐ b) 1
   - ☐ c) 2
   - ☐ d) 4
   - ☐ e) 8

6. What is the value of the dielectric constant of Mica?
   - ☐ a) 1
   - ☐ b) 5.4
   - ☐ c) 12
   - ☐ d) 80
   - ☐ e) 310

7. What is the unit of the drift velocity?
   - ☐ a) $Volt$
   - ☐ b) $Joule$
   - ☐ c) ms^{-1}
   - ☐ d) C
   - ☐ e) F

8. If a 12μF capacitor connected with 12volt source, how much energy stored in the capacitor?
   - ☐ a) 455 Joule
   - ☐ b) 657 Joule
   - ☐ c) 798 Joule
   - ☐ d) 864 Joule
   - ☐ e) 954 Joule

9. Which one is possible?
   - ☐ a) $q = 45e$
   - ☐ b) $q = 35.1e$
   - ☐ c) $q = 25.5e$
   - ☐ d) $q = 15.9e$
   - ☐ e) $q = 5.5e$

10. What is the value of the proportionality constant of Coulomb's law?
   - ☐ a) $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$
   - ☐ b) $9.1 \times 10^{-31} kg$
   - ☐ c) $1.9 \times 10^{-19} C$
   - ☐ d) $1.6 \times 10^{-19} C$
   - ☐ e) $9 \times 10^9 N m^2 C^{-2}$

Invigilator Signature

Examiner Signature